

10. (a) Group 1 and Group 2 metals are in the s-block of the Periodic Table. Using potassium and calcium as examples, discuss the similarities and differences between Group 1 and Group 2 with respect to:

- the reaction of the metals with cold water.
- the solubility of the carbonates.

[6 QER]

You should include appropriate chemical equations in your answer.

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- (b) The equation for the reaction between potassium carbonate and hydrochloric acid is given below.



A 1.40 g sample of impure potassium carbonate was added to excess dilute hydrochloric acid. The impurity is unreactive and only the potassium carbonate reacts with the acid.

The volume of carbon dioxide released was  $186\text{ cm}^3$  when measured at 298 K and  $1.01 \times 10^5\text{ Pa}$ .

Calculate the mass of the impurity.

[3]

mass of impurity = ..... g

- (c) A solution is thought to contain potassium chloride. Describe suitable tests that a student could do to confirm this. Include the expected observations.

[2]

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- (d) By referring to ionisation energies, explain why stable compounds containing  $\text{K}^{2+}$  ions are unlikely to form.

[2]

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